

Exact and Fast Metric Path-Following Coordinates

1 Setup

Let Q be the configuration manifold of the robot and let

$$g_q(v, w) = v^\top M(q)w$$

be the kinetic-energy Riemannian metric. For the two-link example,

$$Q = S^1 \times S^1,$$

with coordinates $q = (\theta_1, \theta_2)$, and $M(q)$ is the manipulator mass matrix.

Let

$$\gamma : [0, 1] \rightarrow Q$$

be a non-self-intersecting reference path. The path-following coordinates are intended to represent a nearby configuration q by

$$(\eta, \xi),$$

where η is the closest point on the path and ξ is the transverse displacement from $\gamma(\eta)$ to q .

2 Exact Theoretical Construction

The exact Riemannian construction uses the geodesic distance induced by g . The coordinate η is defined by

$$\eta = \operatorname{argmin}_{s \in [0, 1]} d_g(\gamma(s), q)^2.$$

Equivalently, if $\log_p^g(q)$ denotes the Riemannian logarithm, then

$$\eta = \operatorname{argmin}_{s \in [0, 1]} \left\| \log_{\gamma(s)}^g(q) \right\|_{g_{\gamma(s)}}^2.$$

The transverse vector is then

$$\xi = \log_{\gamma(\eta)}^g(q).$$

This is the clean geometric object. It has the property that

$$q = \operatorname{Exp}_{\gamma(\eta)}^g(\xi),$$

where Exp^g is the Riemannian exponential map.

2.1 Computational Cost

For a non-constant metric $M(q)$, the logarithm

$$\log_p^g(q)$$

is not a simple coordinate difference. It is obtained by solving the geodesic boundary-value problem

$$\nabla_{\dot{c}} \dot{c} = 0, \quad c(0) = p, \quad c(1) = q.$$

In local coordinates this becomes

$$\ddot{c}^k + \Gamma_{ij}^k(c) \dot{c}^i \dot{c}^j = 0,$$

where the Christoffel symbols Γ_{ij}^k are determined by the metric.

The expensive part is not merely solving one geodesic problem. The closest point search evaluates the objective for many candidate values of s . Thus one controller call may require many BVP solves:

$$s \mapsto \left\| \log_{\gamma(s)}^g(q) \right\|_{g_{\gamma(s)}}^2.$$

This is why the exact implementation is too slow for real-time control.

3 Fast Local Method on a Configuration Submanifold

Now let Q be an arbitrary d -dimensional configuration submanifold, possibly embedded in an ambient vector space $\mathcal{A} \cong \mathbb{R}^m$. The fast method does not require Q to be $S^1 \times S^1$. What it needs is a cheap local replacement for the Riemannian logarithm.

For each $p \in Q$, choose a local retraction

$$R_p : U_p \subset T_p Q \rightarrow Q, \quad R_p(0) = p, \quad DR_p(0) = I.$$

The associated local inverse, when it exists, is denoted

$$\ell_p(q) \approx R_p^{-1}(q) \in T_p Q.$$

This $\ell_p(q)$ is the fast logarithm. It is not the Riemannian logarithm $\log_p^g(q)$; it is a chart or retraction displacement that is cheap to evaluate.

Examples include:

$$\ell_p(q) = \chi_p(q) - \chi_p(p)$$

in a chosen local coordinate chart χ_p , or, for an embedded submanifold, an ambient displacement projected onto $T_p Q$,

$$\ell_p(q) = P_{T_p Q}(q - p),$$

possibly followed by a projection/retraction back to Q . For $S^1 \times S^1$, this is exactly the wrapped joint-space difference.

The fast closest-point coordinate is then

$$\eta_{\text{fast}} = \underset{s \in [0,1]}{\operatorname{argmin}} g_{\gamma(s)}(\ell_{\gamma(s)}(q), \ell_{\gamma(s)}(q)).$$

The fast displacement is

$$\xi_{\text{fast}} = \ell_{\gamma(\eta_{\text{fast}})}(q) \in T_{\gamma(\eta_{\text{fast}})} Q.$$

This keeps the metric g in the objective but replaces the expensive geodesic BVP logarithm by a local algebraic displacement. In a sufficiently small neighborhood of p ,

$$d_g(p, q)^2 = g_p(\ell_p(q), \ell_p(q)) + \text{higher-order terms}.$$

Thus the fast method is the local quadratic model of the Riemannian distance.

4 Lie Group Specialization

If the configuration manifold Q is a Lie group G , there is a natural choice of retraction:

$$R_p(\xi) = p \exp_G(\xi), \quad \ell_p(q) = \log_G(p^{-1}q),$$

where $\exp_G$ and $\log_G$ are the Lie group exponential and logarithm. The corresponding closest-point coordinate is

$$\eta_{\text{Lie}} = \operatorname{argmin}_{s \in [0,1]} g_{\gamma(s)}(\log_G(\gamma(s)^{-1}q), \log_G(\gamma(s)^{-1}q)).$$

Then

$$\xi_{\text{Lie}} = \log_G(\gamma(\eta_{\text{Lie}})^{-1}q).$$

This is still generally not the same as the Riemannian logarithm $\log_p^g(q)$ for the kinetic-energy metric. It becomes the exact Riemannian logarithm only when the chosen Riemannian metric has compatible group invariance. In the two-link example,

$$Q = S^1 \times S^1,$$

and the group is abelian. Therefore

$$\log_G(p^{-1}q) = \text{wrap}(q - p),$$

component by component. This is exactly the same displacement currently used by the fast method. The implementation exposes it as a separate option because the geometric interpretation is different, even though the numerical result is identical for this particular torus.

5 Metric Tangent and Transverse Frame

Let $d = \dim Q$. Along the path, define the metric-unit tangent

$$T(\eta) = \frac{\gamma'(\eta)}{\|\gamma'(\eta)\|_{g_{\gamma(\eta)}}}.$$

Choose a metric-orthonormal transverse frame

$$N(\eta) = [N_1(\eta) \quad \cdots \quad N_{d-1}(\eta)]$$

satisfying

$$g_{\gamma(\eta)}(T, N_a) = 0, \quad g_{\gamma(\eta)}(N_a, N_b) = \delta_{ab}.$$

The transverse coordinate is the metric projection of the fast displacement onto this frame:

$$\zeta = N(\eta)^\flat \xi_{\text{fast}}, \quad \zeta_a = g_{\gamma(\eta)}(\xi_{\text{fast}}, N_a(\eta)).$$

Here $\zeta \in \mathbb{R}^{d-1}$. In matrix coordinates, if $G(\gamma(\eta))$ is the metric matrix and N is the matrix whose columns are N_a , then

$$\zeta = N^\top G(\gamma(\eta)) \xi_{\text{fast}}.$$

For the two-link example $d = 2$, so there is only one normal vector and ζ is the scalar $\xi_\perp$. If

$$\alpha = M(\gamma(\eta))T,$$

then the implementation chooses the metric-normal direction by taking a Euclidean perpendicular to α :

$$\tilde{N} = \begin{bmatrix} -\alpha_2 \\ \alpha_1 \end{bmatrix}, \quad N = \frac{\tilde{N}}{\|\tilde{N}\|_{M(\gamma(\eta))}}.$$

6 Controller

The corresponding local tubular chart is

$$q = \Phi(\eta, \zeta) = R_{\gamma(\eta)}(N(\eta)\zeta), \quad \zeta \in \mathbb{R}^{d-1}.$$

The exact theoretical version replaces the retraction R by the Riemannian exponential Exp^g . The fast implementation uses the same formula but chooses R to be something cheap, such as a wrapped joint chart or an ambient projection/retraction.

Let

$$y = \begin{bmatrix} \eta \\ \zeta \end{bmatrix} \in \mathbb{R}^d.$$

In local coordinates on Q , the acceleration relation is

$$\dot{q} = D\Phi(y)\dot{y}, \quad \ddot{q} = D\Phi(y)\ddot{y} + \frac{d}{dt}(D\Phi(y))\dot{y}.$$

For an embedded representation $q \in \mathcal{A}$, the same equation holds in the ambient space with $\dot{q}, \ddot{q} \in T_q Q$, and $\dot{y}$ can be obtained from a local inverse or metric-weighted pseudoinverse of $D\Phi(y)$. The second term is the path-curvature/centripetal term that is missing from a pure velocity-field controller.

The path coordinate command is

$$\ddot{\eta} = \ddot{\eta}_{\text{ref}} + k_\eta(\dot{\eta}_{\text{ref}} - \dot{\eta}),$$

where $\dot{\eta}_{\text{ref}}$ is chosen so $\|\gamma'(\eta)\dot{\eta}_{\text{ref}}\|_{g_{\gamma(\eta)}}$ equals the desired metric speed. The transverse command is

$$\ddot{\zeta} = -K_p \zeta - K_d \dot{\zeta}.$$

The commanded configuration acceleration is then

$$\ddot{q}_{\text{cmd}} = D\Phi(y) \begin{bmatrix} \ddot{\eta} \\ \ddot{\zeta} \end{bmatrix} + \frac{d}{dt}(D\Phi(y))\dot{y}.$$

Finally, the torque is

$$\tau = M(q)\ddot{q}_{\text{cmd}} + C(q, \dot{q})\dot{q} + G(q).$$

7 Why the Fast Method is Faster

The exact method repeatedly performs:

$$\text{candidate } s \Rightarrow \text{solve geodesic BVP} \Rightarrow \text{evaluate distance.}$$

The fast method performs:

$$\text{candidate } s \Rightarrow \text{local retraction inverse } \ell_{\gamma(s)}(q) \Rightarrow \text{metric quadratic form.}$$

Thus the inner loop changes from nonlinear boundary-value problem solves to simple vector and matrix operations. The approximation is appropriate when the state remains in a sufficiently small tube around a non-self-intersecting path.

8 Limitations

The fast method is local. It should not be interpreted as the exact Riemannian logarithm for large displacements. It requires a computable local chart, projection, or retraction whose inverse is unique in the tube around the path. It also assumes that the path-following tube has a unique closest branch. Self-intersecting paths should be avoided because the coordinate η is not unique at intersections.

For an arbitrary configuration submanifold, the method is therefore available when the following data are available:

$$\gamma(\eta), \quad \gamma'(\eta), \quad g_q, \quad R_p, \quad \ell_p.$$

The robot dynamics do not need to have the special structure of the two-link arm, beyond being fully actuated in the coordinates used for inverse dynamics. The geometry bottleneck is moved from solving geodesic BVPs to evaluating the chosen local retraction model.